

Halogen Bonding between an Isoindoline Nitroxide and 1,4-Diiodotetrafluorobenzene: New Tools and Tectons for Self-Assembling Organic Spin Systems

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Abstract: The isoindoline nitroxide 1,1,3,3-tetramethylisoindolin-2-ylloxyl (TMIO) and 1,4-diiodotetrafluorobenzene readily form a discrete 2:1 complex that shows evidence of relatively strong N–O⋯I halogen bonding. This interaction was characterized in the solid state by single-crystal X-ray analysis, thermal analysis, and vibrational spectroscopy (IR and Raman), backed

by density functional theory calculations. EPR spectroscopy performed on a solution of TMIO in pentafluorobenzene, a halogen-bonding donor, indicates that halogen bonding induces

an increase in electron density at the nitroxide nitrogen nucleus and an increase in the nitroxide rotational correlation time. Our findings demonstrate the potential of utilizing halogen-bonding interactions to promote the self-assembly of new isoindoline nitroxide tectons for the preparation of organic spin systems.

Keywords: halogen bonding • nitroxides • noncovalent interactions • organic spin systems • radicals

Introduction

The controlled self-assembly of molecular species to give well defined supramolecular architectures is an important strategy for the development of functional molecular materials. Although hydrogen bonding and metal coordination have been widely exploited as directing forces in these as-

sembly processes, recent reports have highlighted the efficacy of a less widely utilized interaction—halogen bonding (X-bonding)—for the formation of supramolecular systems.^[1–5] X-bonding encompasses a range of electron donor-acceptor interactions between halogen acceptor atoms (Cl, Br, or I; the X-bonding donors) and electron donors (the X-bonding acceptors), which are typically either electronegative atoms with lone pairs (N, S, O, etc.) or anions. The interaction is noncovalent and shares a number of characteristics with hydrogen bonding, including directionality, strength, and specificity.^[4] Iodoperfluorocarbons form particularly strong X-bonds with acceptors because of the electron-withdrawing inductive effect of the fluorine atoms,^[6–8] and are consequently popular tectons for the formation of supramolecular assemblies.^[1–4,9,10]

While many oxygen- and nitrogen-containing functional groups are known X-bonding acceptors, there have been few investigations of nitroxide radicals in this role, possibly due to the fact that they are generally considered poor electron donors. Recent interest in molecular magnetic materials and emerging fields such as spintronics and quantum computing, however, has inspired the study of X-bonding for directing the self-assembly of nitroxide tectons to create supramolecular arrays with particular spin orientations.^[11] Morishima et al. provided the earliest reports of the interactions between nitroxides and halogenated molecules, de-

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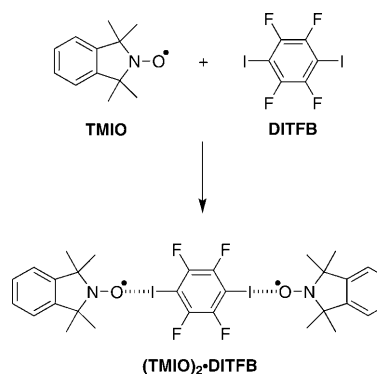
scribing the effect of the radicals on the NMR chemical shifts of halogenated molecules.^[12–14] The interaction was explained in terms of a spin delocalization, or charge transfer interaction, with the assistance of molecular orbital calculations. More recently, complexes of chlorine dioxide with piperidine nitroxides have been reported.^[15,16] These are essentially pre-reactive complexes that spontaneously undergo electron transfer and transform into oxoammonium salts. A crystallographic and magnetic study of 2-halo-substituted nitronyl nitroxides^[17] provided the first direct observation of O⋯I distances and O⋯I–C angles consistent with nitroxide X-bonding, although this term was not used to describe the interaction. While the dominant interactions in the 2-iodo derivative were ferromagnetic, its isomorphous 2-bromo analogue exhibited anti-ferromagnetic behavior. Similarly, one of the two polymorphs of 4-(4-iodobenzylidenamino)TEMPO (TEMPO = 2,2,6,6-tetramethylpiperidin-1-yloxy) was reported to exhibit antiferromagnetic behavior and structural characteristics consistent with X-bonding.^[18] Schöllhorn et al. recently described the first deliberate investigation of nitroxides as X-bond acceptors, reporting the single-crystal X-ray analysis of a one-dimensional co-ordination polymer in which the dominant intermolecular forces are X-bonds involving the nitroxide radical 4-amino-TEMPO.^[11] This report was rapidly followed by an electron paramagnetic resonance (EPR) study of X-bonding between a nitroxide (TEMPO) and aliphatic and aromatic iodoperfluorocarbons in solution.^[19] Notably, the strengths of the observed X-bonds^[11,19] were comparable with the strongest reported in the literature. DFT and MP2 calculations have subsequently been utilized to investigate the physicochemical properties^[20,21] of X-bonded adducts of TEMPO, including interesting deshielding effects consistent with spin transfer observed in the ¹⁹F NMR spectra of TEMPO/haloperfluorocarbon systems.^[21] A recent review of X-bonding in supramolecular chemistry devotes a section to magnetic materials, including nitroxide X-bonded systems.^[22] With the exceptions of NMR studies on acyclic nitroxides^[12–14] and magnetic studies on 2-halo-substituted nitronyl nitroxides,^[17] all of these investigations have involved piperidine-type nitroxides (TEMPO and 4-amino-TEMPO).

Here we chose to investigate the halogen-bonding characteristics of another class of nitroxide radicals: the isoindoline nitroxides. This class of compounds exhibits a number of advantageous characteristics with respect to the piperidine nitroxides, including enhanced stabilities, ease of purification, and superior EPR characteristics.^[23–26] The inherent symmetry, planarity, and rigidity of the fused ring system also results in the formation of well-defined crystals with melting points well above ambient temperatures.^[24,27–31] Such properties recommend this class of stable radicals as potential building blocks for functional molecular spintronic materials.^[32,22] Accordingly, we investigated the simplest stable isoindoline nitroxide—1,1,3,3-tetramethylisoindolin-2-yloxy (TMIO; Scheme 1)^[34]—in the role of an X-bonding acceptor, in view of its symmetry, rigidity, and lack of additional electron donor sites. 1,4-Diiodotetrafluorobenzene (DITFB)

was selected as the X-bonding donor because of its propensity to form well defined X-bonded assemblies with a variety of donor species.^[35–37]

Results and Discussion

Preparation: Well formed yellow crystals were obtained from the slow evaporation of a 2:1 solution of TMIO and DITFB in *n*-hexane (Scheme 1). GC-MS analysis of single crystals in solution indicated the presence of both TMIO and DITFB, which were identified by their mass spectra. Integration of the GC total-ion-count chromatogram supported a 2:1 TMIO/DITFB ratio in the co-crystals, and this molar ratio was verified by analytical microanalysis.



Scheme 1. Formation of the halogen-bonded supramolecular compound (TMIO)₂·DITFB.

Crystallography: Single-crystal X-ray diffraction of the co-crystals confirmed the formation of an X-bonded supramolecular compound that displays crystallographic inversion symmetry and consists of two TMIO radicals and a DITFB molecule: (TMIO)₂·DITFB (Figure 1). The observed O⋯I distance of 2.940 Å is considerably less than the sum of the van der Waals radii (3.50 Å),^[38,39] suggesting a relatively strong X-bonding interaction. This O⋯I distance is slightly longer than that reported for the 4-amino-TEMPO/DITFB complex^[11] (2.827(9) Å), but shorter than that observed for carbonyl-oxygen acceptors with DITFB^[40] or similar donors.^[41] It is shorter than the O⋯X distances both in the homomolecular 4-(iodobenzylidenamino)TEMPO^[18] and in 2-bromo-substituted nitronyl nitroxide^[17] X-bonded systems, but longer than that in the 2-iodo-substituted nitronyl nitroxide^[17] system. The nearly linear O⋯I–C angle (177.6°) is consistent with the n→σ* mode of interaction characteristic of X-bonding. In general, as X-bonds shorten, they tend to be more directional. For example, in the majority of cases, O⋯I–C X-bonds with O⋯I distances less than 3.1 Å have C–I⋯O angles greater than 160°.^[4,42] The combination of angle (177.6°) and distance (2.940 Å) for (TMIO)₂·DITFB is consistent with this relationship. The N–O⋯I angle (137.0°) and C–N–O⋯I torsion angle (8.20°) are consistent with electron donation from the N–O[•] oxygen lone pair, with limited

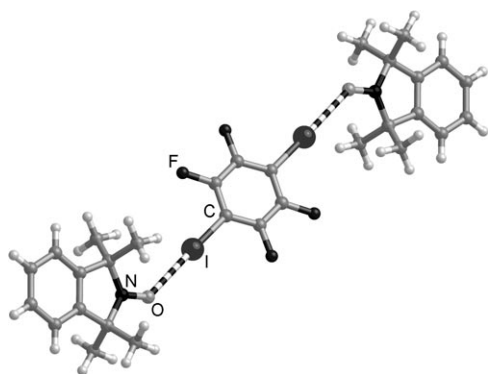


Figure 1. Crystal structure of the supramolecular compound (TMIO)₂·DITFB showing the halogen-bonding interactions between the iodine atoms of DITFB and the nitroxide oxygen atoms of TMIO.

contribution from the π -type arrangement observed for the 4-amino-TEMPO/DITFB complex^[11] and some nitroxides participating in hydrogen bonding.^[43]

There are no significant π - π , or otherwise notable, interactions between the (TMIO)₂·DITFB units, except the ubiquitous and relatively weak dispersion forces (Figure 2). The dominant interaction in the crystal structure therefore appears to be the X-bonding between TMIO and DITFB. It follows that this (TMIO)₂·DITFB system is best described as a supramolecular compound.^[44]

The observed N–O[•] bond length of 1.275 Å in the (TMIO)₂·DITFB co-crystal is considerably longer than those observed in the unit cell of TMIO (1.249(6), 1.252(8)),^[27] although still in the observed range for isoindoline nitroxides (1.249–1.280 Å; s.d. = ± 0.01 Å).^[45] The observed C–I bond length of 2.086 Å is also slightly longer than that observed in pure crystals of DITFB (2.075 Å).^[46]

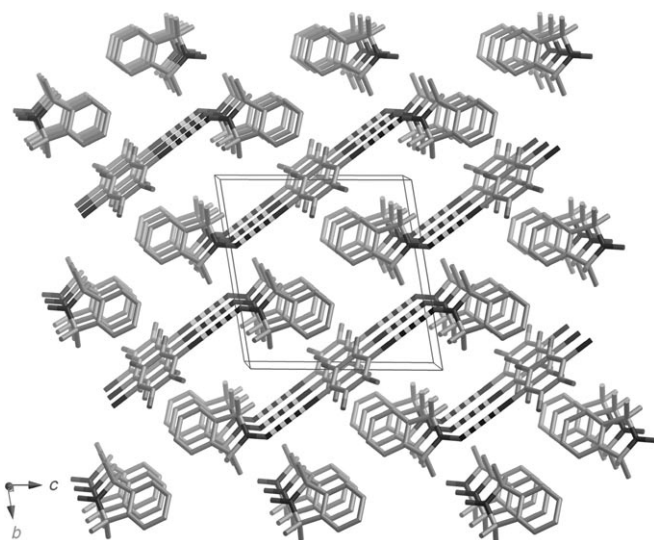


Figure 2. Crystal packing of (TMIO)₂·DITFB. The only significant directional intermolecular interactions are the X bonds in the supramolecular compound.

Thermal analysis: Differential scanning calorimetry (DSC) of the co-crystals and individual components provided insight into the X-bonded system (Figure 3). The heating curves of all three systems exhibited distinct, sharp-melting endotherms, with the melting point (endotherm onset) of the co-crystals (84.3 °C; $\Delta H_{\text{fus}} = 53.4 \text{ kJ mol}^{-1}$) being considerably lower than those of TMIO (127.7 °C; $\Delta H_{\text{fus}} = 26.6 \text{ kJ mol}^{-1}$) and DITFB (106.4 °C; $\Delta H_{\text{fus}} = 21.2 \text{ kJ mol}^{-1}$). In contrast, most reported X-bonded systems exhibit higher melting points than their individual components,^[5,10,40,42,47–49] although it must be noted that these reports describe infinite chains, or networks, held together by halogen-bonding interactions, rather than discrete molecular compounds. Here there are no significant interactions between the (TMIO)₂·DITFB units in the crystal structure, and so a melting point lower than those of the individual components is not unexpected. A similar finding has been reported for the discrete 1:1 X-bonded complexes of imines and *N*-iodo-succinimide.^[50]

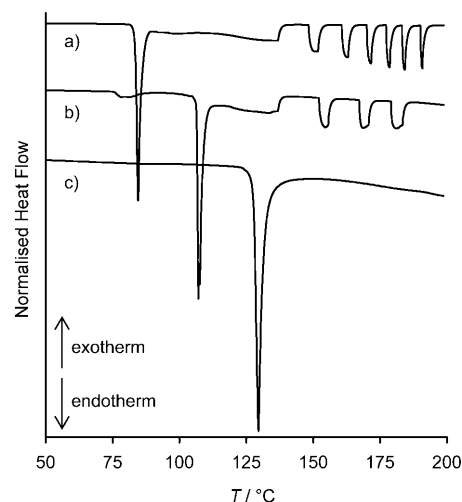


Figure 3. Heating curves of a) (TMIO)₂·DITFB and of the individual components: b) DITFB and c) TMIO.

The series of endotherms in the profile of (TMIO)₂·DITFB at temperatures above 137 °C are indicative of degradative processes. These may be assigned to degradation of the DITFB component on the basis of corresponding features in the thermal profile of pure DITFB. After the first heating cycle, rapid cooling and reheating of the (TMIO)₂·DITFB sample failed to produce further transitions. When a series of heating/cooling cycles (25–100 °C) was performed on (TMIO)₂·DITFB, the only feature observed was the melting endotherm of the first heating sequence, suggesting that an irreversible reaction between the two components occurs at temperatures above the melting point of the complex (but below the melting points of either DITFB or TMIO). This process must be essentially thermally “neutral” because there are no obvious corresponding features in the DSC profile. The formation of an oxoammonium salt through inner-sphere electron transfer from the ni-

troxide to DITFB is a possible outcome. Several authors have described the X-bonding complexes of nitroxides as “pre-reactive” states^[19] and indeed, chlorine dioxide complexes of the piperidine nitroxides TEMPO and 4-hydroxy-2,2,6,6-tetramethylpiperidin-*N*-yloxy (TEMPOL) are known to evolve into the corresponding oxoammonium salts through inner-sphere electron transfer.^[15,16]

Crystalline (TMIO)₂DITFB is stable under normal laboratory conditions and we have not observed any change in this material to date (>1 year). Equimolar solutions of TMIO and DITFB in hexafluorobenzene or toluene were stable for long periods (months). Colorless solutions of TMIO in pentafluoriodobenzene, however, were stable for months in the dark, but yellowed over a period of several days when exposed to visible light.

Vibrational spectroscopy: Interactions between the halogen donor and nitroxide acceptor in the co-crystals were investigated by Raman and infrared spectroscopy, which have been reported to be useful diagnostic probes of halogen bonding.^[51–54] Because the existing reports^[55–57] of calculated vibrational spectra for DITFB do not specify structural assignments of the vibrational modes, ab initio density functional theory (DFT) calculations were performed to obtain unequivocal band assignments for the vibrational frequencies of DITFB (Table 1). The vibrational spectrum of TMIO has previously been analyzed in detail by this approach^[58,59]

with use of the 6–311G+(d) basis set. Although this is the basis set of choice for prediction of vibrational frequencies, it only covers elements up to the third row of the periodic table. To model DITFB, which contains very large iodine nuclei, we employed the well known Lanl2dz basis set, in which the core electrons are treated separately through the use of effective core potentials. The free DITFB molecule belongs to the D_{2h} point group, and the 30 fundamental vibrations are divided among the symmetry species according to $\Gamma_{\text{vib}} = 6A_g + 1B_g + 3B_{2g} + 5B_{3g} + 2A_u + 5B_{1u} + 5B_{2u} + 3B_{3u}$. The DFT predictions are in broad agreement with previous results obtained by the Wilson GF matrix approach,^[55] although many of the symmetry allocations are different. Another minor difference concerns the assignment of the infrared bands at 1356 and 1269 cm^{−1}. These were formerly assigned to a combination band and fundamental (ν_{19}), respectively, whereas DFT suggests the opposite: that the 1356 cm^{−1} band is a fundamental, namely a ring sextant C–C stretch, and that the 1269 cm^{−1} band is a combination mode. In comparison of the DFT predictions for the free molecule with crystal spectra, factor group considerations may apply. DITFB crystals belong to the C_{2h}^5 spectroscopic space group, with two DITFB molecules on general sites per spectroscopic unit cell. Factor group theory predicts that in the crystal spectra each Raman active mode of the free molecule will be reduced in symmetry and potentially split into an A_g and a B_g mode and likewise, each IR active mode will be potentially split into A_u and B_u modes.

Now in possession of reliable assignments of the spectra of the starting materials, we were properly equipped to interpret any spectral changes that occur on co-crystallization of TMIO and DITFB. Although many parts of the IR and Raman spectra of the co-crystal appeared to be little more than simple superpositions of the spectra of the individual components, there are several significant differences that emphasized the presence of distinct TMIO–DITFB interactions (Figures 4 and 5, below). Major differences in the infrared spectra occur in the 1500–1400 cm^{−1} region, where bands generally appear to have changed shape and shifted to lower wavenumbers by 7–10 cm^{−1}. We were particularly interested in the fate of the N–O[•] stretch [$\nu(\text{N–O}^\bullet)$], which appears at 1428 cm^{−1} in the vibrational spectra of polycrystalline

Table 1. Observed and predicted vibrational wavenumbers [cm^{−1}] for DITFB.

	Symmetry	Lanl2dz	Raman	IR	Assignments
1	$\tilde{\nu}_{30}$ B_{3u}	50			
2	$\tilde{\nu}_{27}$ B_{2u}	112			
3	$\tilde{\nu}_{17}$ A_u	123			inactive
4	$\tilde{\nu}_{10}$ B_{2g}	124			
5	$\tilde{\nu}_{15}$ B_{3g}	150			
6	$\tilde{\nu}_6$ A_g	159	156		sym C–I str+ring elongation
7	$\tilde{\nu}_{29}$ B_{3u}	197			
8	$\tilde{\nu}_{22}$ B_{1u}	237			
9	$\tilde{\nu}_5$ A_g	293	306		sym in-plane C–F rock
10	$\tilde{\nu}_{26}$ B_{2u}	300			
11	$\tilde{\nu}_8$ B_{2g}	333	331		out-of-plane ring rock
12	$\tilde{\nu}_7$ B_{1g}	403	401		out-of-plane ring twist
13	$\tilde{\nu}_{21}$ B_{1u}	416			
14	$\tilde{\nu}_{14}$ B_{3g}	438	439		in-plane ring def
15	$\tilde{\nu}_4$ A_g	488	498		lateral ring expansion+(C–I+C–F) sym str
16	$\tilde{\nu}_{28}$ B_{3u}	616		563	ring deformation out-of-plane
17	$\tilde{\nu}_{16}$ A_u	675	665		inactive
18	$\tilde{\nu}_{13}$ B_{3g}	696	705		
19	$\tilde{\nu}_{20}$ B_{1u}	733		759	C–I antisym str
20	$\tilde{\nu}_8$ B_{2g}	793	807		ring deformation out-of-plane
21	$\tilde{\nu}_3$ A_g	863	875		C–I sym str
22	$\tilde{\nu}_{25}$ B_{2u}	912		942	C–F side-to-side stretch
23	$\tilde{\nu}_{12}$ B_{3g}	1095	1133		C–F diagonal stretch
24	$\tilde{\nu}_{19}$ B_{1u}	1190		1214	in-plane ring bending by sextant
				1269	combination band
25	$\tilde{\nu}_{24}$ B_{2u}	1339		1356	aryl sextant C–C str
26	$\tilde{\nu}_2$ A_g	1373	1383		symmetric ring contraction and F extension
27	$\tilde{\nu}_{18}$ B_{1u}	1426		1430	aryl semicircle stretch
28	$\tilde{\nu}_{23}$ B_{2u}	1460		1460	aryl semicircle stretch
29	$\tilde{\nu}_{11}$ B_{3g}	1605	1576		aryl quadrant stretch
30	$\tilde{\nu}_1$ A_g	1631	1609		aryl quadrant stretch

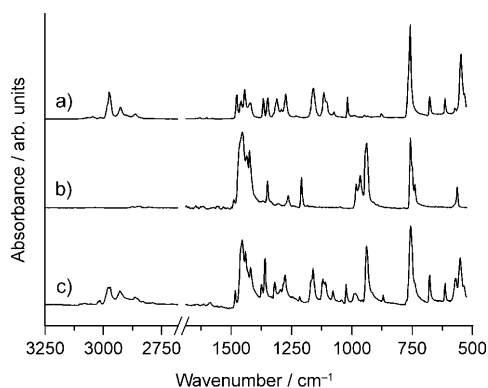


Figure 4. Fingerprint region of the IR spectra (ATR, diamond cell) of a) crystalline TMIO, b) DITFB, and c) (TMIO)₂·DITFB co-crystal.

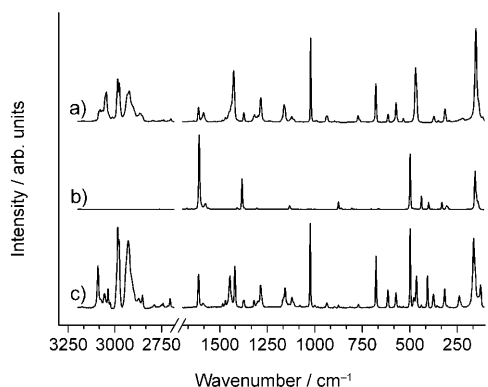


Figure 5. Fingerprint region of the Raman spectra of a) crystalline TMIO, b) DITFB and, c) (TMIO)₂·DITFB co-crystal.

TMIO.^[58,59] Although there is a sharp band at 1418 cm^{-1} in the IR spectrum of the co-crystal, it is unclear whether this is due to $\tilde{\nu}(\text{N}-\text{O}^{\cdot})$ or to the semicircle ring stretch of the DITFB component, which in pure DITFB occurs at 1430 cm^{-1} . More extensive changes in this region were observed in the Raman spectrum (Figure 5), in which the strong 1426 cm^{-1} band in the TMIO spectrum [$\tilde{\nu}(\text{N}-\text{O}^{\cdot})$] and the strong sharp band at 1383 cm^{-1} (a simultaneous ring contraction and fluorine extension) in the DITFB spectrum seemed to have no direct analogues in the spectrum of the co-crystal. Instead two strong sharp bands appear at 1445 and 1420 cm^{-1} , so the exact fate of $\tilde{\nu}(\text{N}-\text{O}^{\cdot})$ remains unclear. Attempts to obtain assignments for the co-crystal by DFT calculations have up to this point proved unsuccessful, due to the time penalty resulting from the large size of the complex (72 atoms), the large degrees of freedom inherent in such a complex, and also possibly the inability of the Lanl2dz basis set to model the $\text{N}-\text{O}^{\cdot}\cdots\text{I}$ interaction suitably.

A number of other vibrational modes of TMIO are affected by halogen bonding, although it is possible that some of the differences between the spectra of TMIO and (TMIO)₂·DITFB are due to crystal field effects. As one example, in the IR spectrum the $\tilde{\nu}(\text{C}-\text{N})$ absorption at 1280 cm^{-1} in TMIO appears to have shifted to 1277 cm^{-1} in the complex, whereas it barely shifts in the Raman spec-

trum. The Raman-active isindoline ring elongation and ring backbone flex bands also shift to lower and higher wavenumbers, respectively. There are, of course, two TMIO moieties in the unit cell of the co-crystal, related by a center of symmetry. Each TMIO mode in the co-crystal is thus split into an in-phase Raman-active vibration and an out-of-phase IR-active counterpart, and so care must be taken in the interpretation of subtle shifts.

On going from a free to an X-bonded system, the $\tilde{\nu}(\text{C}-\text{H})$ bands of X-bonding acceptors are reported to exhibit shifts to higher wavenumber,^[5,42,49] and decrease in intensity with respect to the fingerprint region of the spectrum.^[5,42] Here however, the IR and Raman $\tilde{\nu}(\text{CH}_3)$ bands ($2800\text{--}3000\text{ cm}^{-1}$) of TMIO show relatively small shifts ($<6\text{ cm}^{-1}$) upon X-bonding with DITFB. While this may be a consequence of X-bonding, it is difficult to discount the influence of steric effects in the new structure. Similarly, the Raman $\tilde{\nu}(\text{CH}_3)$ region for TMIO exhibits an increase in relative intensity with respect to the fingerprint region upon X-bonding. The Raman $\tilde{\nu}(\text{aryl C}-\text{H})$ region ($3040\text{--}3100\text{ cm}^{-1}$) is also significantly altered. Again, it is difficult to assign these effects to X-bonding, particularly in view of the sensitivity of relative Raman band intensities to orientation effects in crystalline samples.

The majority of changes in the spectra of DITFB upon interaction with TMIO are consistent with previous reports of DITFB involvement in X-bonding.^[5,40,42] The IR-active aryl quadrant stretch (1461 cm^{-1}), C-F side-to-side stretch (942 cm^{-1}), and C-I antisymmetrical stretch (759 cm^{-1}) of DITFB undergo shifts ($<7\text{ cm}^{-1}$) to lower wavenumbers, although the first and last of these overlap with TMIO bands. DFT calculations assigned the Raman-active band at 156 cm^{-1} to a combined symmetrical C-I stretch and ring elongation and the 498 cm^{-1} band to a lateral ring expansion in conjunction with C-I and C-F symmetrical stretches. Previous reports^[5,49] indicate that " $\tilde{\nu}(\text{C}-\text{I})$ " bands typically shift to lower wavenumber, although here the 156 cm^{-1} band moves to higher wavenumber (164 cm^{-1}) and the 498 cm^{-1} band remains essentially unchanged. Most of the DITFB Raman bands, with the exception of the 498 cm^{-1} band, are considerably reduced in intensity in the (TMIO)₂·DITFB spectrum, with some bands, including the strong aryl ring contraction and F-extension band at 1383 cm^{-1} , completely disappearing.

EPR spectroscopy: A previous report by Lucarini et al.^[19] described perturbation of the EPR spectrum of TEMPO in solution by the presence of X-bonding donors. This was manifested primarily as an increase in the isotropic nitrogen hyperfine coupling (a_N), with a minimal effect on the g value (g_{iso}). In iodoperfluorocarbon solvents, the EPR spectra of diphenylnitroxide were markedly broadened. Here, we utilized EPR spectroscopy to probe the effect of X-bonding on the nitroxide moiety of TMIO in solution, comparing the EPR spectrum of TMIO in hexafluorobenzene to that obtained in pentafluoriodobenzene, an X-bond donor (Figure 6). Notably, the spectrum of TMIO in hexafluoro-

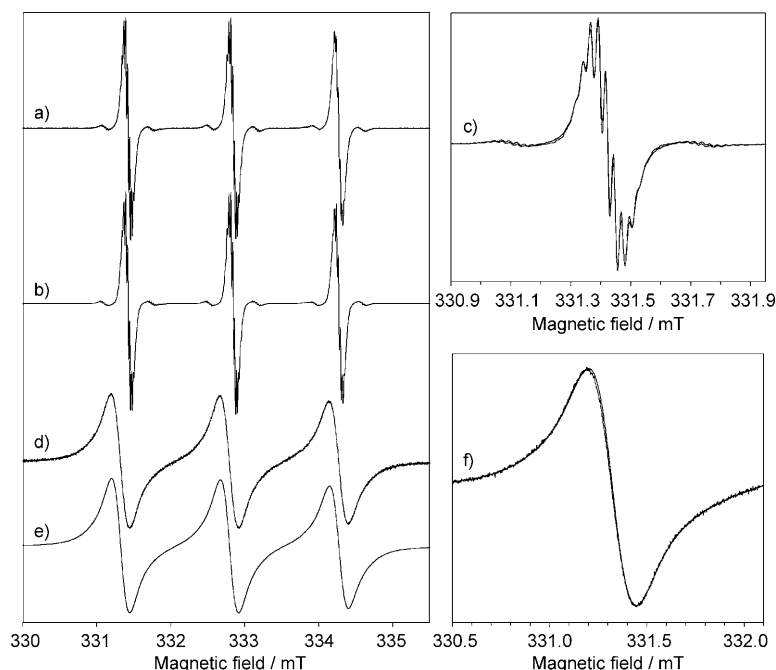


Figure 6. a) Experimentally measured, and b) simulated EPR spectra of TMIO (50 μM) in hexafluorobenzene with c) corresponding overlaid experimental and simulated low-field manifolds showing well-resolved proton hyperfine coupling ($\nu=9.34499$ GHz; $T=298$ K). d) Experimentally measured, and e) simulated EPR spectra of TMIO (50 μM) in pentafluoroiodobenzene, with f) corresponding overlaid experimental and simulated low-field manifolds exhibiting extensive broadening due to halogen-bonding interactions ($\nu=9.34465$ GHz; $T=298$ K).

benzene exhibited remarkably narrow line widths, with well resolved isotropic nitrogen (^{14}N , $I=1$, 99.632 % abundant, ^{15}N , $I=1/2$, 0.368 % abundant), carbon (^{13}C , $I=1/2$, 1.07 % abundant) and proton (^1H , $I=1/2$, 99.985 % abundant; ^2H , $I=1$, 0.015 % abundant) hyperfine couplings, the latter arising from the four magnetically equivalent methyl groups. Computer simulation (spectrum b, Figure 6) of the experimental spectrum (spectrum a, Figure 6) was performed with the XSophe-Sophe-XeprView computer simulation software suite^[60] and employed the following spin Hamiltonian:

$$H = g_{\text{iso}}\beta\mathbf{B} \cdot \underline{\mathbf{S}} + \alpha_N \underline{\mathbf{S}} \cdot \underline{\mathbf{I}}_N - g_{n(N)}\beta_n \underline{\mathbf{B}} \cdot \underline{\mathbf{I}}_N + \sum_{i=1}^4 \alpha_C \underline{\mathbf{S}} \cdot \underline{\mathbf{I}}_C - g_{n(C)}\beta_n \underline{\mathbf{B}} \cdot \underline{\mathbf{I}}_C + \sum_{i=1}^{12} \alpha_H \underline{\mathbf{S}} \cdot \underline{\mathbf{I}}_H - g_{n(H)}\beta_n \underline{\mathbf{B}} \cdot \underline{\mathbf{I}}_H \quad (1)$$

and Kivelson's line width model^[61–65]

$$\sigma_{M_I} = \alpha + \beta M_I + \gamma M_I^2 + \delta M_I^3 \quad (2)$$

$$\alpha = \alpha' + \alpha''$$

where α'' arises from a spin rotational mechanism and α' , β , γ , and δ arise from the modulation of the g and A matrices.^[61–65] The spin Hamiltonian [$g_{\text{iso}}=2.00597$, $a_N=13.33$, $a_C=6.004$, $a_H=0.235 \times 10^{-4} \text{ cm}^{-1}$ ($a_N/g_{\text{iso}}\beta=1.4234$, $a_C/g_{\text{iso}}\beta=0.6411$, $a_H/g_{\text{iso}}\beta=0.0251 \text{ mT}$)], and line width ($\alpha=0.186$, $\beta=-0.004$, $\gamma=0.010$, $\delta=-0.002 \times 10^{-4} \text{ cm}^{-1}$) parameters were optimized by use of a quadratic version of the Hooke and

Jeeves optimization algorithm.^[60] While the line shape function is over-parameterized (four variables vs. three line widths), it is clear that the α term dominates, indicating that in hexafluorobenzene the g and A anisotropy is almost completely averaged through rapid tumbling of TMIO. Under normal circumstances proton hyperfine coupling is unresolved in EPR spectra of nitroxides such as TMIO and TEMPO, being observed only under conditions under which line-broadening due both to electronic relaxation and to motional anisotropy is minimized (e.g., 233 K in deoxygenated toluene for TMIO).^[26] In contrast, the spectrum obtained in the X-bonding donor pentafluoroiodobenzene (spectrum d, Figure 6) exhibited a dramatic increase in line width,

with a concomitant reduction in signal intensity. Computer simulation of this spectrum (spectrum e, Figure 6), employing an approach similar to that used above for TMIO in hexafluorobenzene, produced the spin Hamiltonian [$g_{\text{iso}}=2.00619$, $a_N=13.81$, $a_C=6.004$, $a_H=0.235 \times 10^{-4} \text{ cm}^{-1}$ ($a_N/g_{\text{iso}}\beta=1.4234$, $a_C/g_{\text{iso}}\beta=0.6411$, $a_H/g_{\text{iso}}\beta=0.0251 \text{ mT}$)], and line width ($\alpha=2.0143$, $\beta=-0.0459$, $\gamma=0.0403$, $\delta=0.000132 \times 10^{-4} \text{ cm}^{-1}$) parameters. Unresolved ^{13}C and ^1H hyperfine couplings were left unchanged in optimization of the spin Hamiltonian parameters. Again, the line width is dominated by α . The origin of the line broadening may result either from unresolved iodine (^{127}I ; $I=5/2$) hyperfine coupling or an increase in the rotational correlation time, a consequence of the increased hydrodynamic radius of the TMIO–pentafluoroiodobenzene X-bonded species. Whereas α' is proportional to the rotational correlation time [Eq. (3)],

$$\alpha' = \tau_c \left[\frac{4}{45} (\Delta\gamma H_0)^2 + \frac{3}{40} I(I+1)b^2 \right]$$

$$\alpha'' = (\Delta g_{\parallel}^2 + \Delta g_{\perp}^2) kT / 12\pi\rho R^3$$

where

$$\tau_c = 4\pi\rho R^3 / 3kT \quad (3)$$

$$\Delta\gamma = \beta\Delta g / \hbar$$

$$b = -2/3(A_{\parallel} - A_{\perp})$$

$$\Delta g = g_{\parallel} - g_{\perp}$$

$$g_{\parallel} = g_{\parallel} - 2.00231$$

$$g_{\perp} = g_{\perp} - 2.00231$$

α'' is inversely proportional to the rotational correlation time (τ_c) and should not contribute significantly to α if the hydrodynamic radius (R) is increased. Estimation of the hydrodynamic radii of TMIO (6.78 Å) and TMIO-pentafluoriodobenzene (15 Å) from the crystal structure of (TMIO)₂-DITFB and using the expression for α' and the value of α (0.186 mT), under the assumption that the g - and A -anisotropy remains constant, produces a value of $\alpha' = 2.01$ mT, which is very similar to the value of α (2.0134 mT) determined experimentally. This would suggest that the line-broadening arises from a larger rotational correlation time, a consequence of an increased hydrodynamic radius of the TMIO-pentafluoriodobenzene X-bonded species. If this is correct, then the delocalization of the nitroxide unpaired electron onto the iodine atom (^{127}I ; $I = 5/2$) of the solvent through X-bonding interactions must be small. The isotropic nitrogen hyperfine interaction increased slightly from $13.33 \times 10^{-4} \text{ cm}^{-1}$ to $13.81 \times 10^{-4} \text{ cm}^{-1}$, which is consistent with an increase in spin density at the nitrogen nucleus of the nitroxide, because the ionic resonance structure of TMIO becomes slightly more stable through the halogen bonding of the N–O[•] moiety with the iodine nucleus.^[20] The isotropic g -value was also slightly affected, increasing from 2.00597 to 2.00619. Lucarini and co-workers described similar effects for TEMPO in the presence of X-bond donors.^[19]

The EPR spectra of 1:1 mixtures of TMIO and DITFB were recorded both in toluene and in hexafluorobenzene at ambient temperature (298 K) and in toluene at low temperature (130 K), and compared with those of TMIO under the same conditions. Interestingly, it was not possible to distinguish between the spectra of the radical in the presence or in the absence of the equimolar X-bonding donor, indicating that the TMIO X-bonding interaction is fairly weak in solution.

Conclusions

We have demonstrated that isoindoline nitroxides are effective electron donors for the formation of halogen bonds with iodoperfluorocarbons. The formation of a discrete halogen-bonded supramolecular compound from TMIO and DITFB—(TMIO)₂-DITFB—in the solid phase was conclusively demonstrated by single-crystal X-ray crystallography, with the geometry about the N–O[•]···I interaction consistent with $n \rightarrow \sigma^*$ electron donation from the nitroxide N–O[•] moiety. Vibrational spectroscopy, supported by DFT calculations, provided further insight into the halogen-bonding interactions in the solid state. In solution, the effect of halogen bonding on the nitroxide moiety was evaluated by EPR spectroscopy, which provided evidence of a slight increase in electron density at the nitroxide nitrogen nucleus as a consequence of the stabilization of the ionic resonance structure of the TMIO nitroxide radical through halogen bonding of the iodine atom. The strength and directionality of the halogen bonding observed here suggest the use of isoindoline nitroxides and iodoperfluorocarbons as building blocks for the

assembly of supramolecular arrays of paramagnetic molecules. Further insight into the characteristics and control of these interactions should assist in the design and synthesis of functional molecular materials with potential application in the fields of molecular magnetism, spintronics, and quantum computing.

Experimental Section

General: 1,1,3,3-Tetramethylisoindolin-2-yloxy (TMIO) was synthesized by the procedure of Griffiths et al.^[34] 1,4-Diiodotetrafluorobenzene (DITFB) and pentafluoriodobenzene were purchased from Sigma–Aldrich and used as obtained. DSC thermal analysis was performed on a TA Instruments DSC Q2000 instrument in standard mode over temperature ranges of 25–200 °C or 25–100 °C with a heating rate of 10 °C min^{−1} and a continuous nitrogen purge of 50 cm³ min^{−1}.

(TMIO)₂-DITFB: Co-crystals were obtained by slow evaporation of an n -hexane (1 cm³) solution of TMIO (10.0 mg, 52.6 μmol) and DITFB (10.6 mg, 26.3 μmol, 0.5 equiv), with essentially quantitative mass recovery: $R_f = 0.48$ (TMIO), 0.60 (DITFB) (EtOAc/hexane 3:7); m.p. 83–85 °C; IR (ATR): $\tilde{\nu}_{\text{max}} = 2980, 2971, 2929, 1454, 1440, 1419, 1360, 1277, 1161, 939, 757, 678, 551 \text{ cm}^{-1}$; Raman (solid): $\tilde{\nu}_{\text{max}} = 3091, 2985, 2978, 2927, 1612, 1446, 1420, 1025, 678, 498, 465, 407, 164 \text{ cm}^{-1}$; elemental analysis calcd (%) for C₃₀H₃₂F₄I₂N₂O₂: C 46.38, H 4.22, N 3.58; found: C 46.05, H 4.12, N 3.58.

GC-MS analysis of single co-crystals gave two peaks identified by their mass spectra as TMIO and DITFB. TMIO: MS (EI⁺): m/z (%): 190 (35) [M]⁺, 175 (24), 160 (88), 145 (100). DITFB: MS (EI⁺): m/z (%): 402 (100) [M]⁺, 275 (24) [$M-I$]⁺, 148 (85) [C_6F_4]⁺, 98 (37), 79 (26).

Spectroscopy: Infrared spectra were obtained by use of a horizontal, single bounce, diamond ATR accessory on a Nicolet Nexus 870 FT-IR instrument. Spectra were recorded for 32 scans at 4 cm^{−1} resolution with an OPD velocity of 0.6289 cm s^{−1}. Solids were pressed directly onto the diamond internal reflection element of the ATR without further sample preparation. Raman spectra were obtained either on a Renishaw System 1000 Raman microscope with use of a HeNe laser giving 10 mW of 633 nm radiation at the sample, or on a Renishaw inVia Raman microscope fitted either with a 532 nm doubled, diode pumped Nd-YAG laser or with a 785 nm laser diode, each giving up to 100 mW at the sample. Raman spectra were recorded with resolutions of better than 4 cm^{−1} in all cases. A quartz wedge depolarizer was inserted in the excitation beam path to reduce unwanted polarization effects in crystalline samples. Calibration was performed by referencing the spectrometer to the 520.5 cm^{−1} band of a silicon wafer.

EPR spectra were recorded on a Bruker Elexsys E580 EPR spectrometer (X-band, ≈ 9.2 GHz) with use of a Bruker microwave frequency counter and a Bruker ER036M teslameter for microwave frequency and magnetic field calibration. TMIO solutions (50 μM) in hexafluorobenzene and pentafluoriodobenzene were thoroughly deoxygenated by bubbling with argon immediately before analysis. Spectra were recorded at 298 K. TMIO and 1:1 TMIO/DITFB solutions (1 mM in nitroxide) were also prepared in toluene and hexafluorobenzene, and EPR spectra were recorded at 130 K (toluene only) and 298 K. EPR spectra were simulated by use of the XSophe-Sophe-XeprView computer simulation software suite^[60] running on a personal computer with Mandriva Linux 2008 as the operating system.

Quantum chemistry calculations: Calculations were performed with the Gaussian 03 program^[66] and the GaussView3.0 (Gaussian, Inc., Wallingford, CT) front end, running on an SGI Origin 3000 supercomputer with 116 processors and 60 GB of memory. The vibrational spectrum of DITFB was calculated by density functional theory (DFT) by the RB3-LYP method with the LanL2dz basis set. No scaling factor was applied to the calculated wavenumbers.

X-ray crystallography: Crystal data for (TMIO)₂-DITFB were collected with a Bruker SMART 1000 diffractometer and graphite-monochromated

Table 2. Crystal data and details of the structure determination for (TMIO)₂-DITFB.

formula	C ₃₀ H ₃₂ F ₄ I ₂ N ₂ O ₂
molecular weight	782.38
crystal size [mm]	0.50 × 0.44 × 0.21
crystal color	yellow
crystal habit	prism
crystal system	triclinic
space group	<i>P</i> $\bar{1}$
<i>a</i> [Å]	5.8172(8)
<i>b</i> [Å]	11.8006(16)
<i>c</i> [Å]	11.8175(16)
α [Å]	79.636(2)
β [Å]	76.859(2)
γ [Å]	78.675(2)
<i>V</i> [Å ³]	766.79(18)
ρ_{calcd} [g cm ⁻³]	1.694
<i>Z</i>	1
<i>T</i> [K]	150(2)
$\lambda(\text{MoK}\alpha)$ [Å]	0.71073
$\mu(\text{MoK}\alpha)$ [mm ⁻¹]	2.104
<i>T</i> (empirical) _{min,max}	0.534, 0.643
$2\theta_{\text{max}}$ [°]	56.82
<i>hkl</i> range	−7–7, −15–15, −15–15
<i>N</i> (<i>N</i> _{var})	9957 (185)
<i>N</i> _{ind} (<i>R</i> _{merge})	3628 (0.0193)
<i>N</i> _{obs} [<i>I</i> > 2σ(<i>I</i>)]	3518
<i>R</i> ^[a] [<i>I</i> > 2σ(<i>I</i>)], <i>wR</i> ^[a] (all)	0.0168, 0.0400
GoF	1.015
residual extrema [e Å ⁻³]	−0.555, 0.373

[a] $R1 = \sum ||F_o| - |F_c|| / \sum |F_o|$ for $F_o > 2\sigma(F_o)$ and $wR2 = [\sum [w(F_o^2 - F_c^2)^2] / \sum [w(F_c^2)^2]]^{1/2}$ where $w = 1/[\sigma^2(F_o^2) + (0.010P)^2 + 0.75P]$, $P = (F_o^2 + 2F_c^2)/3$.

Mo-K α radiation generated from a sealed tube. Data integration and reduction were undertaken with SAINT and XPREP^[67] and subsequent computations were carried out by use of the WinGX-32^[68] graphical user interface. The structures were solved by direct methods with SIR97^[69] and were then refined and extended with SHELXL-97.^[70] A multi-scan empirical absorption correction was applied with SADABS.^[71] Non-hydrogen atoms were refined anisotropically. Hydrogen atoms were included in idealized positions and refined with a riding model. Crystal data and refinement details are provided in Table 2, and an ORTEP^[72] depiction of the structure is supplied as Supporting Information.

CCDC 699475 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

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